

DISASTER MANAGEMENT

GEOPHYSICAL DISASTERS

MODULE - 5



DISASTER MANAGEMENT

DISASTERS ARISING FROM NATURAL HAZARDS

GEOPHYSICAL

Geological process or phenomenon that may cause loss of life, injury or other health impacts, property damage.



HYDROLOGICAL

Events caused by deviations in the normal water cycle and/or overflow of bodies of water caused by wind set-up.



METEOROLOGICAL

Events caused by short-lived/small to meso-scale atmospheric processes - in the spectrum from minutes to days.



CLIMATOLOGICAL

Events caused by long-lived meso- to macro-scale processes (intra seasonal to multi-decadal).



BIOLOGICAL

Events that have organic origin or conveyed by biological vectors, pathogens, toxins and bioactive substances



DISASTER MANAGEMENT

DISASTERS ARISING FROM NATURAL HAZARDS

GEOPHYSICAL

- Earthquakes
- Volcano
- Tsunami



HYDROLOGICAL

- Avalanche
- Coastal Erosion
- Coastal Flood
- Hydrological flood
- Flash Flood
- Debris/mud flow



METEOROLOGICAL

- Cyclone
- Storm Surge
- Heat wave/ cold wave
- Heavy rains, Sand or dust storm



CLIMATOLOGICAL

- Droughts
- Extreme hot or cold conditions
- Forest fires
- Glacial lake outburst



BIOLOGICAL

- Epidemics – viral, bacterial, parasitic, fungal or prion infections
- Insect infestations
- Animal stampedes



GEOFYSICAL HAZARDS

EARTHQUAKE

Any seismic event—whether natural or caused by humans—that generates seismic waves and cause movement of soil.



VOLCANO

A rupture in the crust, that allows hot lava, volcanic ash, and gases to escape from a magma chamber below the surface.



TSUNAMI

Tsunamis are caused by earthquakes or volcanic eruptions under the sea which results in giant waves near the shore



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GEOPHYSICAL HAZARD

EARTHQUAKE



SECONDARY DISASTERS

- Landslide following earthquake;
- Urban fires triggered by earthquakes;
- Liquefaction - the transformation of (partially) water-saturated soil from a solid state to a liquid state caused by an earthquake
- Mass movement of earth materials, usually down slopes
- Surface displacement of earthen materials due to ground shaking triggered by earthquakes

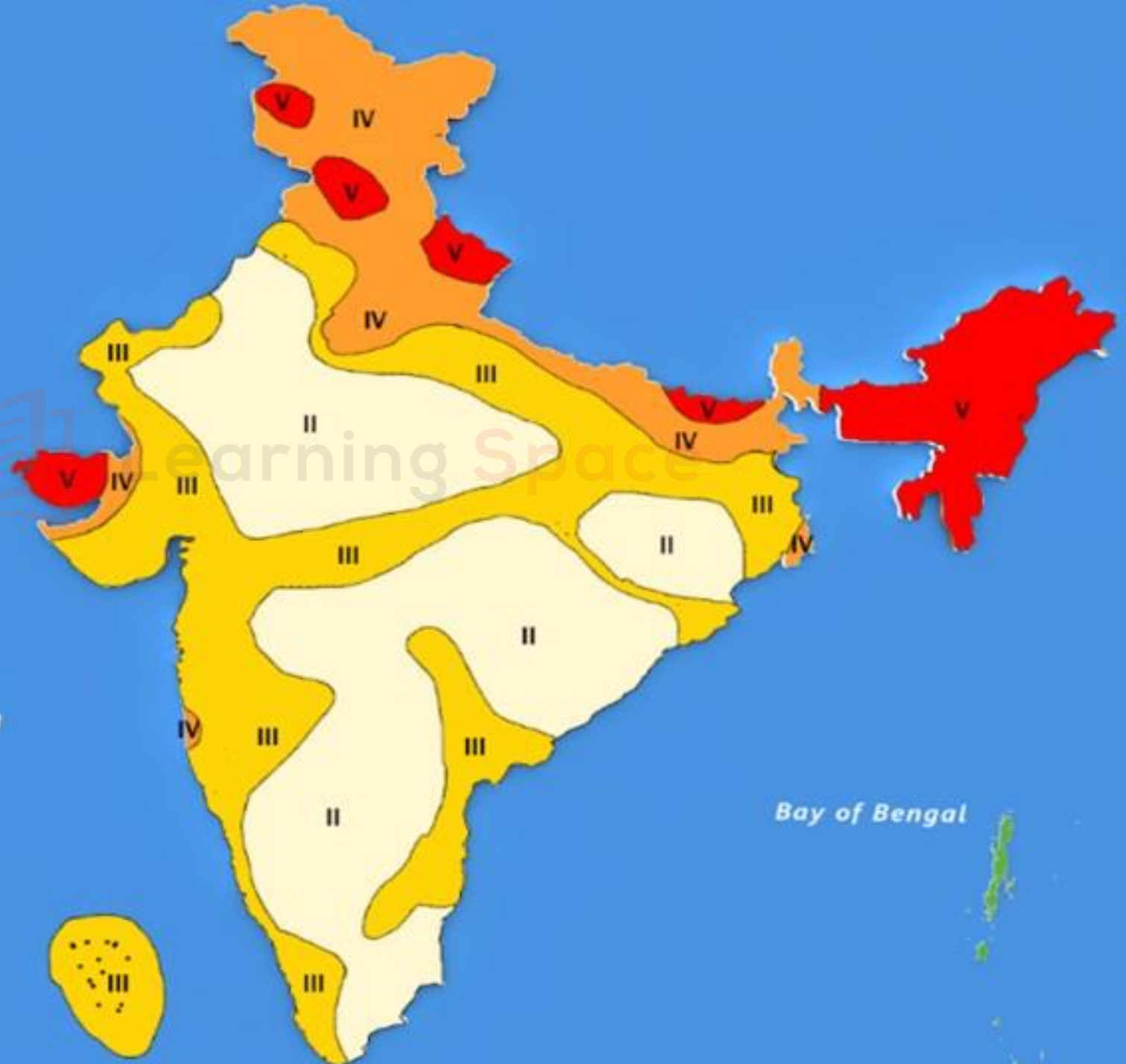


EARTHQUAKE HAZARD ZONES IN INDIA

ZONE	INTENSITY
Zone V	Very High Risk Zone Area liable to shaking Intensity IX (and above)
Zone IV	High Risk Zone Intensity VIII
Zone III	Moderate Risk Zone Intensity VII
Zone II	Low Risk Zone VI (and lower)

Arabian Sea

Bay of Bengal



UNDERSTANDING DISASTER RISK

- Estimating earthquake parameters by National Seismological Network, Real Time Seismic Monitoring Network (RTSMN), Earthquake Hazard and Risk Assessment (EHRA)
- Scientific seismic zonation (Zone II, III, IV, V)
- Conduct micro zonations, prioritize important urban areas for micro-zonation



INTERAGENCY COORDINATION

- Overall disaster governance by MOES
- Response (Organising and coordinating central assistance) – MHA
- Non-structural measures - MOES, MHA, BIS, NDMA
- Coordination among state agencies for non structural measures.



INVESTING IN DRR – STRUCTURAL MEASURES

- Ensuring multi-hazard resistant features are incorporated into Social Housing Schemes
- Strengthening and seismic retrofitting of prioritized lifeline structures and buildings
- Hazard resistant construction, strengthening, and retrofitting of all lifeline structures and critical infrastructure



INVESTING IN DRR – NON STRUCTURAL MEASURES

- Periodic updates on regulations and model codes for town planning, civil works and public infrastructure
- Structural safety audit of lifeline structures and buildings
- Prioritization of lifeline structures and buildings for strengthening and seismic retrofitting
- Establish a professional Civil Engineers Council established by an Act for certification of engineers and evolve a procedure for certification of engineers



CAPACITY DEVELOPMENT

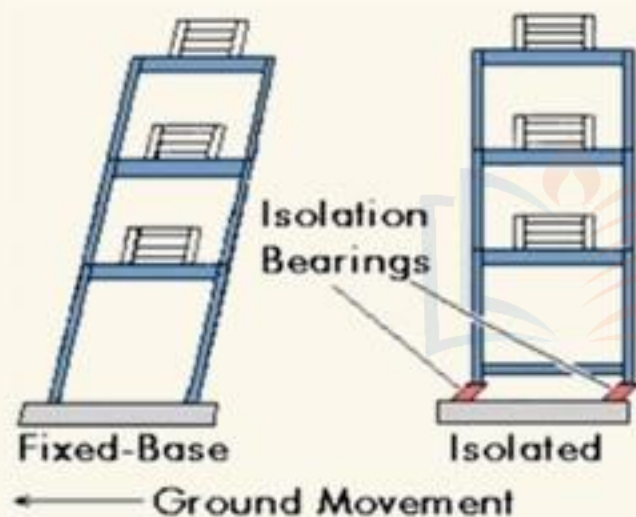
- Facilitate the introduction of subjects related to DM, in the undergraduate and professional courses
- Promote a general awareness, attitude and behaviour change in the awareness campaigns
- Promote use of insurance/ risk transfer
- Promote mock drills, exercises, emergency drills by NDRF, Armed Forces, CAPF and even to the local organisations.
- Empowering women, marginalised, and persons with disabilities with guidance.



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SEISMIC RISK REDUCTION

RETROFITTING BUILDINGS



- The key idea of making a building earthquake-resistant is to make it ductile, i.e. to give it a certain flexibility to shake horizontally.
- It helps soften the impact of the earthquake and lets the building absorb its energy.
- To make a building earthquake resistant, its base is strengthened in a way that during an earthquake, the building's load is borne by the base alone, and upper stories do not experience much quaking.

- The part of the base that is above the ground is cut and rested on bearings.
- The bearings act as shock absorbers, similar to those in cars.
- Adding rubber material such as used tyres to the foundation of a building under construction can also be done.
- In India, the Building Materials & Technology Promotion Council (BMTPC) has undertaken projects for retrofitting of life-line structures.



ALSO KNOW – 'EARTHQUAKE SWARM' IN PALGHAR



1 Palghar has been witnessing an unusual frequency of earthquakes since November, 2018. National Centre for Seismology (NCS) confirmed that when seismic energy piles up inside the Earth and is released in small amounts from certain points, such a series of earthquakes can occur

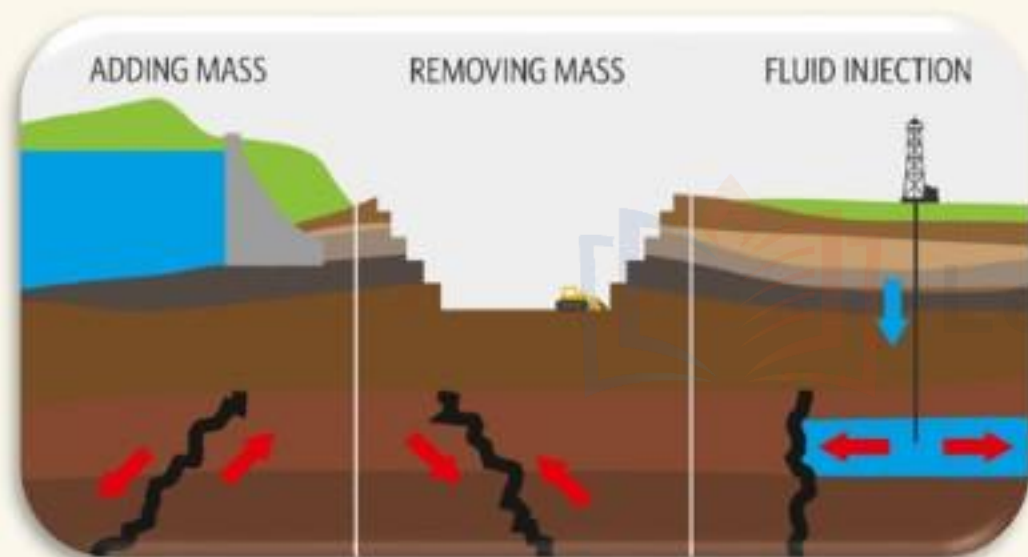
2 NCS has categorised the unusual tremors as an 'earthquake swarm' which is a series of low magnitude earthquakes that occur in a localised region and over a period of time ranging from days, weeks to even months.

3 But these 'swarms' also can come as foreshocks to the main earthquake which could be much greater in magnitude. Two such instances were the Latur and Koyna earthquakes in 1993 and 1967 respectively killing thousands of people.

4 Apart from Palghar, Gujarat's Saurashtra and Kutch regions have also experienced repeated low intensity earthquake shocks which were caused due to intra-plate seismicity.



ALSO KNOW – HUMAN INDUCED SEISMIC ACTIVITY



1 Oil and gas extraction using fluid injection, as well as wastewater disposal, is known to increase seismicity rate in surrounding regions.

2 A new study has found that injection of fluid into subsurface of the earth (one kilometer deep) can cause events like earthquakes and can result in earthquakes spread over larger regions, going far beyond the area invaded by the injected fluids.

3 Understanding this science behind fluid-induced earthquakes could help in unraveling reservoir-induced earthquakes in Koyna, Maharashtra.

4 The study is done to yield data about fault behaviour at greater depths in the earth's crust so that more reliable models of earthquake hazard can be produced.



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GEOPHYSICAL HAZARD

VOLCANO

The geological event near an opening/vent in the Earth's surface including volcanic eruptions of lava, ash, hot vapour, gas, and pyroclastic material



Barren Islands volcano, India's only active volcano, is reportedly being active after a gap of 150 years

SECONDARY DISASTERS

- Surface displacement of earthen materials due to ground shaking triggered by volcanic eruptions
- Ash fall; Lahar - Hot or cold mixture of earthen material flowing on the slope of a volcano either during or between volcanic eruptions;
- Lava Flow
- Pyroclastic Flow - Extremely hot gases, ash, and other materials of more than 1,000 degrees Celsius that rapidly flow down the flank of a volcano (more than 700 km/h) during an eruption.



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GEOPHYSICAL HAZARD

TSUNAMI

SECONDARY DISASTERS



- Tsunamis are difficult to categorize they are essentially an oceanic process that is manifested as a coastal water-related hazard.
- A series of waves with long wavelengths when traveling across the deep ocean that are generated by a displacement of massive amounts of water through underwater earthquakes, volcanic eruptions or landslides.
- Tsunami waves travel at very high speed across the ocean but as they begin to reach towards land, slow down and the wave grows in size causing widespread destruction.



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TSUNAMI RISK REDUCTION

UNDERSTANDING DISASTER RISK

- Development of large scale digital maps indicating tsunami hazard from past events.
- Prepare database of Tsunami risk and vulnerability
- Securing critical instrumentation for early warning
- Providing warning by monitoring seismic activities
- Studies on vulnerabilities and capacities covering social, physical, economic, ecological, gender, social inclusion and equity aspects

INTERAGENCY COORDINATION

- Overall disaster governance by MOES
- Response (Organising and coordinating central assistance) - MHA
- Warnings, Information, Data - MOES, NDMA
- Non-structural measures - MOES, MHA, BIS, NDMA
- SDMA, DDMA, Panchayats - Ensuring coherence and mutual reinforcement of DRR, CCA and development in the coastal areas

INVESTING IN DRR – STRUCTURAL MEASURES

- Strengthening of lifeline structures and high priority buildings
- Inland dykes to safeguard vital installations
- Hazard resistant construction
- Strengthening, and retrofitting of all lifeline structures and critical infrastructure
- Periodical dredging of the inlets and associated water bodies so as to absorb the influx during tsunami



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TSUNAMI RISK REDUCTION

INVESTING IN DRR – NON STRUCTURAL METHODS

- Mainstreaming DM into Development Planning
- Implementation and popularization of relevant Indian Standards
- Support State Government in preparing bye-laws for rural areas (for both engineered and nonengineered buildings)
- Using non-structural shore stabilization measures and bio-shields
- Safety audits and evaluation of all lifeline structures and important facilities
- Investing in private-public partnerships.

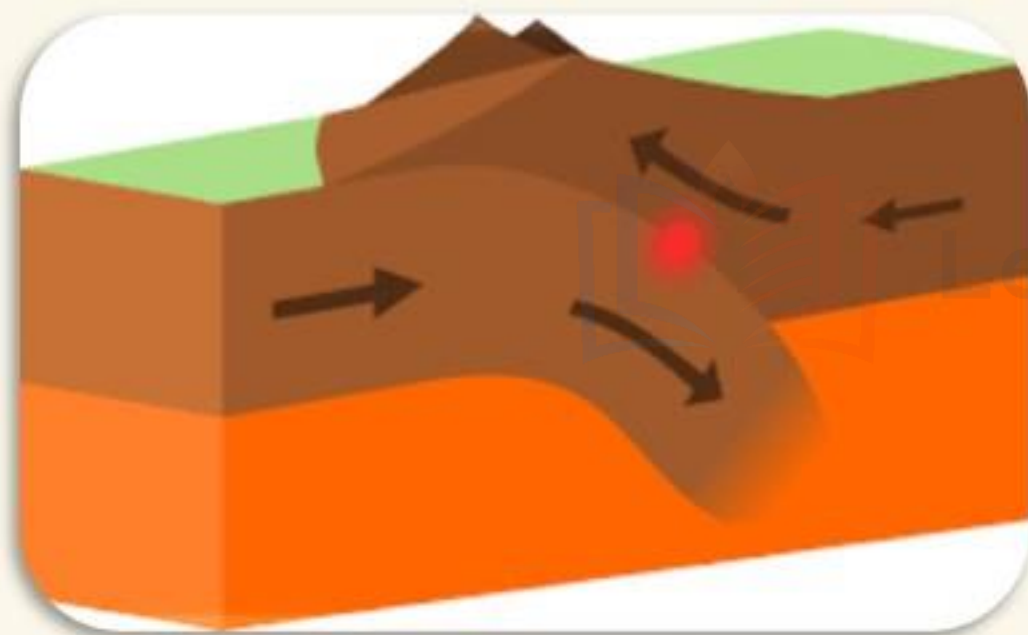


CAPACITY DEVELOPMENT

- Conduct training programmes for State and Local Administration personnel including Fire and Rescue and Police personal in disaster management
- Comprehensive curriculum for tsunami management in the form of training modules for the various target groups
- Carry out mass media campaigns mainly for a attitude change
- Joint planning and execution of emergency drills (Central and State.
- Incorporating gender sensitive and equitable approaches in capacity development



ALSO KNOW – WHAT ARE TSUNAMIGENIC ZONES?



1 Tsunamigenic zones are the regions in the sea where vertical shifting of water mass can cause Tsunami waves.

2 Region of convergent tectonic plate boundaries are the most likely sources for Tsunami

3 Two main Indian Ocean Tsunamigenic zones are Makran subduction zone (MSZ) in the Northern Arabian Sea and Indonesian subduction zone (ISZ) in the Bay of Bengal near Indonesian Islands.

4 MSZ in the north west and ISZ in the south east are not far from India's coast and a major earthquake can be disastrous for India.

5 2004 Tsunami was generated in the ISZ, which is very near to the Andaman and Nicobar Islands and about 1300km from the mainland India



GREENLAND

Barents Sea

TSUNAMIGENIC ZONES NEAR INDIAN COASTS

RUSSIA

MONGOLIA

CHINA

INDIA

North Pacific Ocean

North Atlantic Ocean

Makran
subduction zone
(MSZ)

Indonesian
subduction
zone (ISZ)

AUSTRALIA

Indian Ocean

South Atlantic Ocean

BRAZIL



LESSONS LEARNT FROM PAST TSUNAMI DISASTER



- 1
 - The Indian Tsunami Early Warning Centre (ITEWC) established at Indian National Centre for Ocean Information Sciences, (INCOIS - ESSO) Hyderabad
 - It is integrated with the system of other Asian countries.
 - Multilateral exercises conducted as IOWAVE, BLUE WAVE

- 2
 - While the system can send out warnings quickly, dissemination of the alerts to people on the ground takes time.
 - Coastal infrastructure is still poor and gaps needs to be filled with awareness and community preparedness activities
- 3
 - Mangroves require special attention from government.
 - Traditional knowledge of the Andaman and Nicobar tribal population and scientific discourses should be integrated
 - Only a very handful of scientists are working on Tsunami research today.





RELATED QUESTIONS THAT CAN BE ANSWERED AFTER READING THE MODULE



- 1** What are the types of natural hazards? How can understanding the risk help in disaster risk reduction. Examine.
- 2** List out NMDA guidelines for effective disaster risk reduction in the occasion of an earthquake.
- 3** What is retrofitting? Why is it done to some buildings? How retrofitting buildings can prove resilience during any disaster?
- 4** What are Tsunamigenic zones? How are they situated in the vicinity of Indian coastal region. Examine the preparedness level of India to minimize the impact of Tsunamis



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